Artifical Intelligence Basics Course Summary

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1 Key Terms

1.1 Artificial Neural Networks

- **Perceptron**: A fundamental unit in neural networks, consisting of weighted inputs, a summation node, and an activation function that produces an output based on the weighted sum of inputs and a threshold.

1.2 Search Algorithms

- **Minimax Search**: A decision-making algorithm used in game playing, where the algorithm searches the game tree to find the optimal move for the current player, assuming the opponent will also play optimally.
- **Alpha-Beta Pruning**: An optimization technique used to reduce the search space in Minimax search by eliminating branches that cannot affect the optimal decision.

1.3 Machine Learning Algorithms

- **Reinforcement Learning**: A type of machine learning where an agent learns to make decisions by interacting with an environment and receiving rewards or penalties based on its actions.
- **Inductive Reasoning**: The process of deriving general principles from specific observations.
- **Linear Regression**: A machine learning algorithm used to model the relationship between a dependent variable and one or more independent variables by fitting a linear equation to observed data.
- **Model Evaluation**: The process of assessing the performance of a machine learning model.
- **Overfitting**: A phenomenon in machine learning where a model learns the training data too well, resulting in poor generalization to unseen data.
- **Empirical Risk Minimizer**: A machine learning approach that aims to find the model that minimizes the average loss on the training data.
- **Decision Boundary**: A line or surface that separates different classes of data points in a classification problem.
- **0-1 Loss**: A loss function in classification that assigns a loss of 1 if the prediction is incorrect and 0 if it is correct.
- **Perceptron Loss**: A loss function used in the perceptron algorithm, defined as max\0, -z\, where z is the margin.
- **Hinge Loss**: A loss function used in support vector machines (SVMs), defined as max\0, 1 - z\, where z is the margin.
- **Logistic Loss**: A loss function used in logistic regression, defined as log(1 + e^{-z}), where z is the margin.
- **Stochastic Gradient Descent**: An optimization algorithm that updates model parameters using the gradient calculated from a randomly selected subset of the training data.
- **Logistic Regression**: A machine learning algorithm used for binary classification that models the probability of a data point belonging to a particular class using a logistic function.

1.4 Linear Algebra

- **Matrix**: A rectangular array of numbers, symbols, or expressions, arranged in rows and columns.

- **Linear Transform**: A function that maps vectors from one vector space to another, satisfying $T(u+v) = T(u) + T(v)$ and $T(cu) = cT(u)$ for all vectors u, v and scalars c.
- **Matrix Multiplication**: An operation that combines two matrices to produce a third matrix, where the element at row i and column j of the resulting matrix is the dot product of row i of the first matrix and column j of the second matrix. The inner dimensions must match.
- **Matrix Inverse**: For a square matrix A, its inverse A^{-1} is a matrix such that $AA^{-1} = A^{-1}A = I$, where I is the identity matrix.
- **Matrix Transposition**: An operation that swaps the rows and columns of a matrix. The transpose of matrix A is denoted as A^T.
- **Derivative**: A measure of how a function changes with respect to a change in its input.
- **Partial Derivative**: A derivative of a function of multiple variables with respect to one of those variables, holding the others constant.
- **Sigmoid Function**: A mathematical function that maps any input value to an output value between 0 and 1. Often used as an activation function in neural networks.

1.5 Philosophy of AI

- **Turing Test**: An operational test for intelligent behavior, where a human evaluator engages in natural language conversations with both a human and a machine, attempting to distinguish them. The machine passes if the evaluator cannot reliably identify it as such.
- **Chinese Room Argument**: A thought experiment that challenges the validity of the Turing Test as a measure of true intelligence, arguing that a system can pass the test without possessing genuine understanding.

2 Problem Types

2.1 AI Fundamentals

- **Understanding the Nature of Intelligence**: This problem explores the definition and characteristics of intelligence, both human and artificial.
- **Defining Artificial Intelligence**: This problem involves formulating a precise definition of artificial intelligence, considering various perspectives and historical contexts.
- **Evaluating Intelligent Behavior**: This problem focuses on developing methods for assessing whether a machine exhibits intelligent behavior, including the Turing Test and its limitations.
- **Overcoming the Limitations of Search**: This problem addresses the computational challenges of exhaustive search in complex games, highlighting the need for machine learning techniques.
- **Exploring the Similarities Between AI Tasks**: This problem focuses on identifying commonalities between seemingly disparate AI tasks (e.g., speech synthesis and game playing) in terms of algorithm design and problem-solving approaches.
- **Understanding the AI Stack**: This problem involves understanding the different layers and sub-fields within the broader field of Artificial Intelligence and how they interact.

2.2 Game AI

- **Understanding the Differences Between Game Types**: This problem involves comparing and contrasting different game types (e.g., chess vs. soccer) in terms of their complexity and the challenges they pose for AI.
- **Understanding the Components of Game Playing Systems**: This problem involves understanding the different components of game playing systems, including knowledge representation, action selection, and opponent modeling.
- **Understanding AlphaGo's Architecture**: This problem involves understanding the architecture of AlphaGo, a successful AI system for playing Go, and how it combines deep learning and search techniques.

2.3 Machine Learning Algorithms

- **Developing Algorithms for Complex Tasks**: This problem involves designing algorithms for complex tasks, considering various components such as knowledge representation, decision-making, and generalization.
- **Understanding Perceptrons**: This problem involves understanding the structure and function of perceptrons, a fundamental building block of neural networks.
- **Understanding Reinforcement Learning**: This problem involves understanding the principles of reinforcement learning and how it can be applied to game playing.
- **Linear Regression**: Finding the line that best fits a set of data points, minimizing the sum of squared distances between the points and the line.
- **Multiple Linear Regression**: Extending linear regression to include multiple independent variables to predict a dependent variable.
- **Curve Fitting**: Fitting a curve to a set of data points, potentially using polynomial functions of higher degrees.
- **Binary Classification**: Classifying data points into two categories (+1 or -1) using a linear model and a suitable loss function.
- **Overfitting**: The phenomenon where a model fits the training data too closely, resulting in poor generalization to unseen data.
- **Choosing a Loss Function**: Selecting an appropriate loss function (e.g., 0-1 loss, perceptron loss, hinge loss, logistic loss) for binary classification, considering differentiability and computational efficiency.
- **Minimizing Loss using Gradient Descent**: Employing gradient descent or stochastic gradient descent to iteratively update model parameters and minimize the chosen loss function.

2.4 Linear Algebra for AI

- **Representing Linear Transformations with Matrices**: Matrices are used to represent linear transformations, which map vectors from one space to another while preserving linear combinations. This is fundamental to many AI algorithms.
- **Matrix Arithmetic**: The slide covers basic matrix operations such as addition, subtraction, scalar multiplication, and multiplication, which are essential for many AI computations.
- **Solving Systems of Linear Equations**: Systems of linear equations can be represented and solved using matrices and their inverses. This is crucial for various AI applications.
- **Matrix Transposition**: Matrix transposition is a fundamental operation that involves swapping rows and columns of a matrix, useful for data manipulation in AI.
- **Parallelizing Matrix Multiplication**: The slide demonstrates how matrix multiplication can be parallelized using domain decomposition, improving computational efficiency in AI.

2.5 Calculus for AI

- **Calculus in AI**: The slide highlights the importance of calculus, specifically derivatives, partial derivatives, and finding maxima/minima/gradients, in AI algorithms.
- **Calculating the Derivative of the Sigmoid Function**: The slide shows the step-by-step derivation of the derivative of the sigmoid function using calculus rules.

2.6 AI Safety and Ethics

- **Mitigating AI Safety Risks**: This problem investigates the potential risks and vulnerabilities of AI systems, such as adversarial attacks and biases.
- **Addressing Ethical Concerns in AI**: This problem examines the ethical implications of AI systems, particularly concerning bias and fairness.

2.7 Computer Vision

- **Addressing the Challenges of Computer Vision**: This problem explores the difficulties and ongoing research in computer vision, contrasting initial optimism with the persistent complexities.

3 Algorithm Solutions

3.1 Linear Algebra

- **Matrix Addition**: $A + B = C$, where $c_{ij} = a_{ij} + b_{ij}$
- **Matrix Scalar Multiplication**: $k \cdot A = B$, where $b_{ij} = k \cdot a_{ij}$
- **Matrix Multiplication**: $C_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$
- **Matrix Transposition**: Swapping rows and columns of a matrix A to obtain A^T.
- **Solving Linear Equations with Invertible Matrices**: $Ax = b \implies x = A^{-1}b$
- **Parallel Matrix Multiplication using Domain Decomposition**: Dividing the matrix multiplication task into sub-tasks that can be processed concurrently on multiple cores.

3.2 Artificial Neural Networks

- **Derivative of Sigmoid Function**: $\frac{d}{dx}\sigma(x) = \frac{e^{-x}}{(1 + e^{-x})^2}$
- **Perceptron Algorithm**: Iteratively update the weight vector w based on misclassified data points. If $sign(w^T x_n) \neq y_n$, then $w \leftarrow w + y_n x_n$.

3.3 Machine Learning Algorithms

3.4 Search Algorithms

- **Minimax Search**: A search algorithm for two-player zero-sum games that aims to find the best move for the current player, assuming the opponent also plays optimally. It uses an evaluation function to assign scores to game states and recursively explores the game tree, alternating between maximizing the score for the current player and minimizing the score for the opponent.
- **Alpha-Beta Pruning**: A technique to reduce the search space in Minimax search by avoiding the evaluation of branches that cannot affect the optimal decision. It uses alpha (best value for the maximizing player) and beta (best value for the minimizing player) values to prune branches.